Experiment No. 9

Aim: Create an ERC 20 Token in Remix IDE

Theory:

ERC-20 (Ethereum Request for Comments 20) is a standard for creating and managing fungible tokens on the Ethereum blockchain. Fungible tokens are interchangeable, meaning that each token of the same type has identical properties and can be exchanged for another. ERC-20 provides a common set of rules that allows tokens to interact with wallets, decentralized applications, and cryptocurrency exchanges that support the standard.

ERC-20 tokens are implemented using smart contracts written in Solidity. A smart contract defines the token's behaviour, including its total supply, account balances, transfers, and permissions for spending tokens. Once deployed on the blockchain, the contract executes transactions according to its programmed rules.

The ERC-20 standard defines several important functions. The totalSupply() function returns the total number of tokens in circulation, while balanceOf() returns the number of tokens held by a particular account. The transfer() function allows an account to send tokens to another account. The approve() function allows an account owner to authorize another account to spend a specified number of tokens. The allowance() function checks the remaining approved amount, and transferFrom() enables an authorized spender to transfer tokens on behalf of the owner.

The standard also defines two important events: Transfer and Approval. The Transfer event records token movements between accounts, while the Approval event records changes in spending permissions. These events make it easier for applications and users to track token-related activities.

Output:

Tutorial 1:

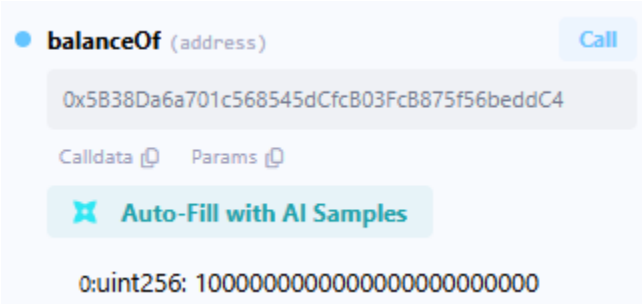
```
1   // SPDX-License-Identifier: MIT
2   pragma solidity ^0.8.3;
3
4   contract Quiz {
5       uint8 public question1 = 0;
6       uint8 public question2 = 0;
7       uint8 public question3 = 0;
8   }
```

Tutorial 2:

```
1   // SPDX-License-Identifier: MIT
2   pragma solidity ^0.8.3;
3
4   import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";
5
6   contract MyToken is ERC20 {
7       constructor() ERC20("MyToken", "MTK") {    infinite gas 968200 gas
8           _mint(msg.sender, 1000000 * 10 ** decimals());
9       }
10  }
```

```
[vm] from: 0x5B3...eddC4 to: MyToken.approve(address,uint256) 0xd91...39138 value: 0 wei data: 0x095...00014 logs: 1 hash: 0x93c...f267c   Debug

status                  1 Transaction mined and execution completed
transaction hash        0x93c800be48fc751f721058f4e678e0b792e3591db4a8fd137f69015366ff267c
block hash              0xae255dbd5226356380aba29ce4e88731f080885d5c772e67eb08fbcf3525b68c
```

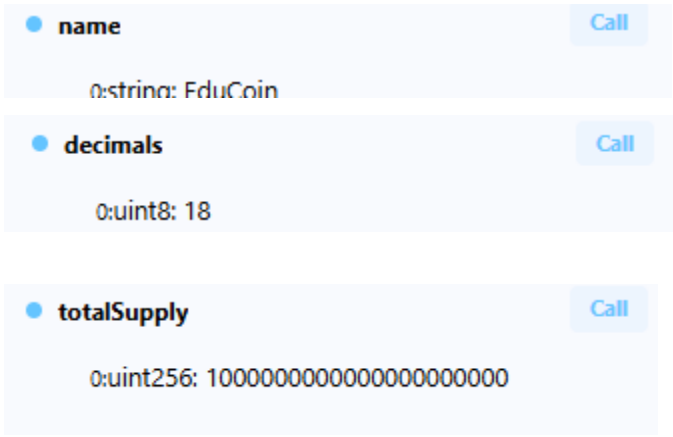


Tutorial 3:

```
1   // SPDX-License-Identifier: MIT
2   pragma solidity ^0.8.3;
3
4   import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";
5
6   contract EduCoin is ERC20 {
7       constructor() ERC20("EduCoin", "EDC") {    infinite gas 968200 gas
8           _mint(msg.sender, 1000 * 10 ** decimals());
9       }
10  }
```

```
[vm] from: 0x5B3...eddC4 to: EduCoin.approve(address,uint256) 0xf8e...9fBe8 value: 0 wei data: 0x095...000c8 logs: 1 hash: 0x067...0f47c   Debug

status                  1 Transaction mined and execution completed
transaction hash        0x067baf0cd0439eec38088fc026d5160d352ba9cd3f5056ae238fbb495030f47c
block hash              0x2488d1258dab3e6fb4714f13def4b6b8e7e85846dc7ffa43146fad7df7284be6
block number            4
```

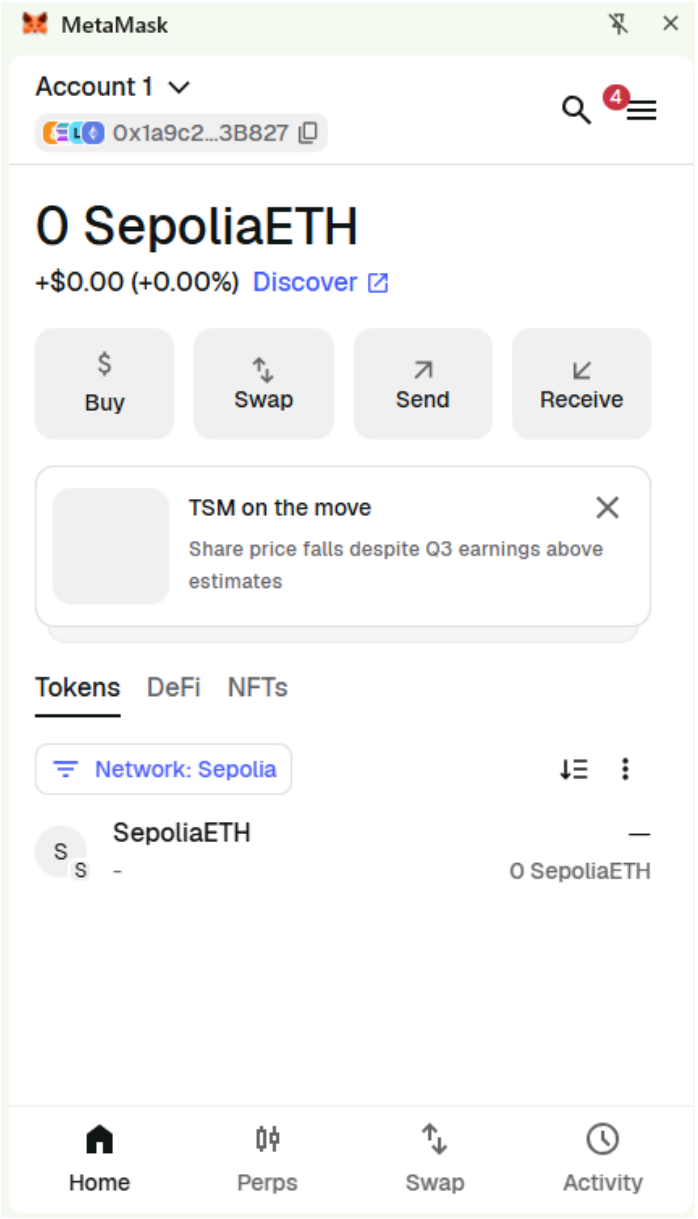


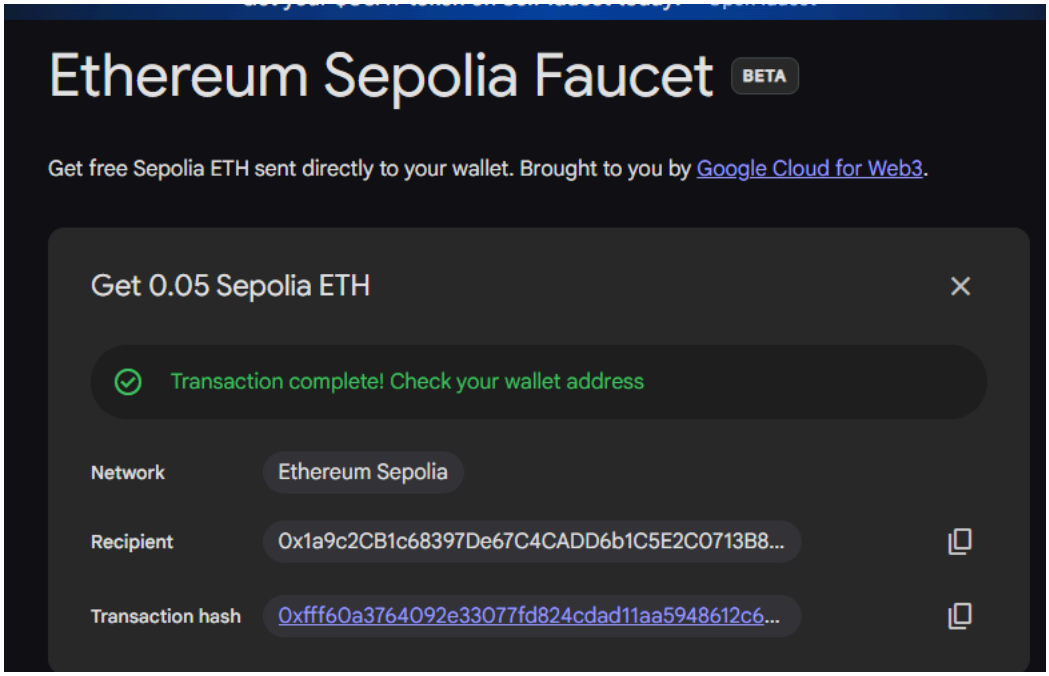
Tutorial 4:

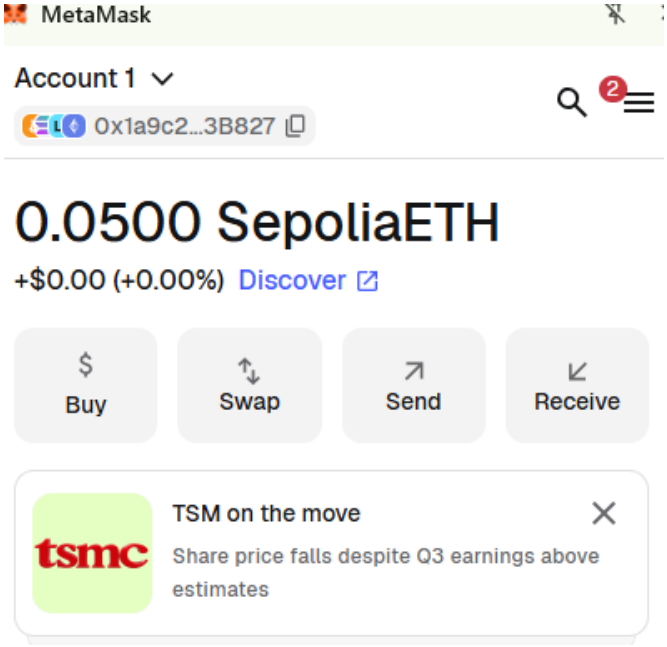
```solidity
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;

import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";

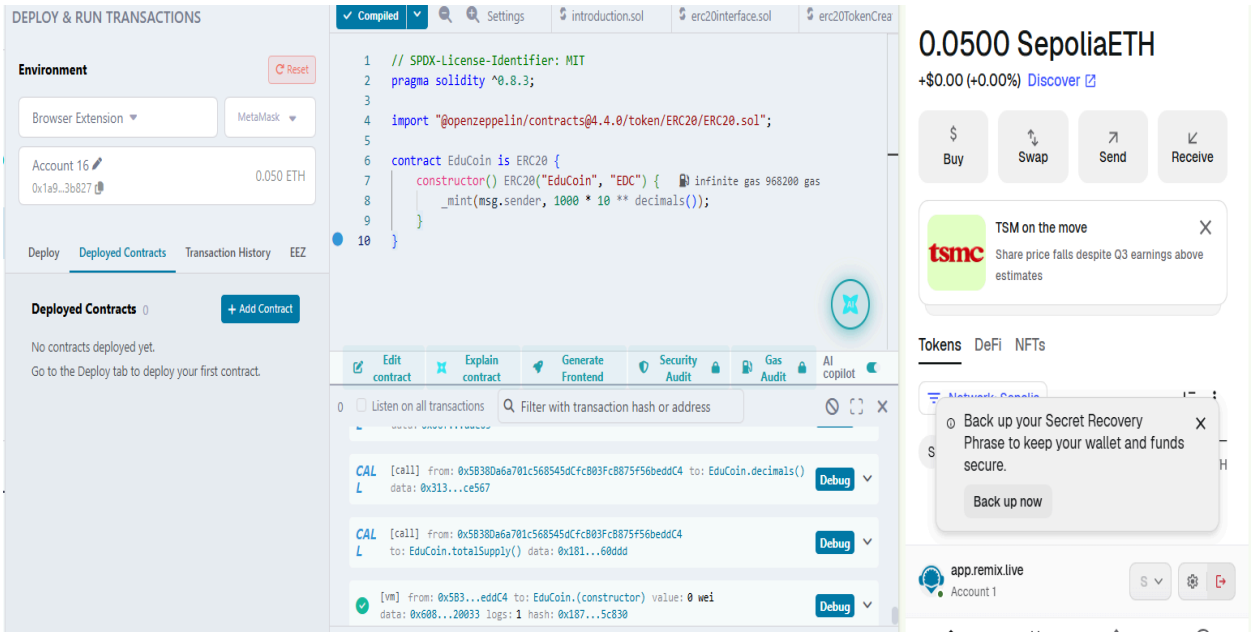
contract EduCoin is ERC20 {
    constructor() ERC20("EduCoin", "EDC") {    infinite gas 968200 gas
        _mint(msg.sender, 1000 * 10 ** decimals());
    }
}
```

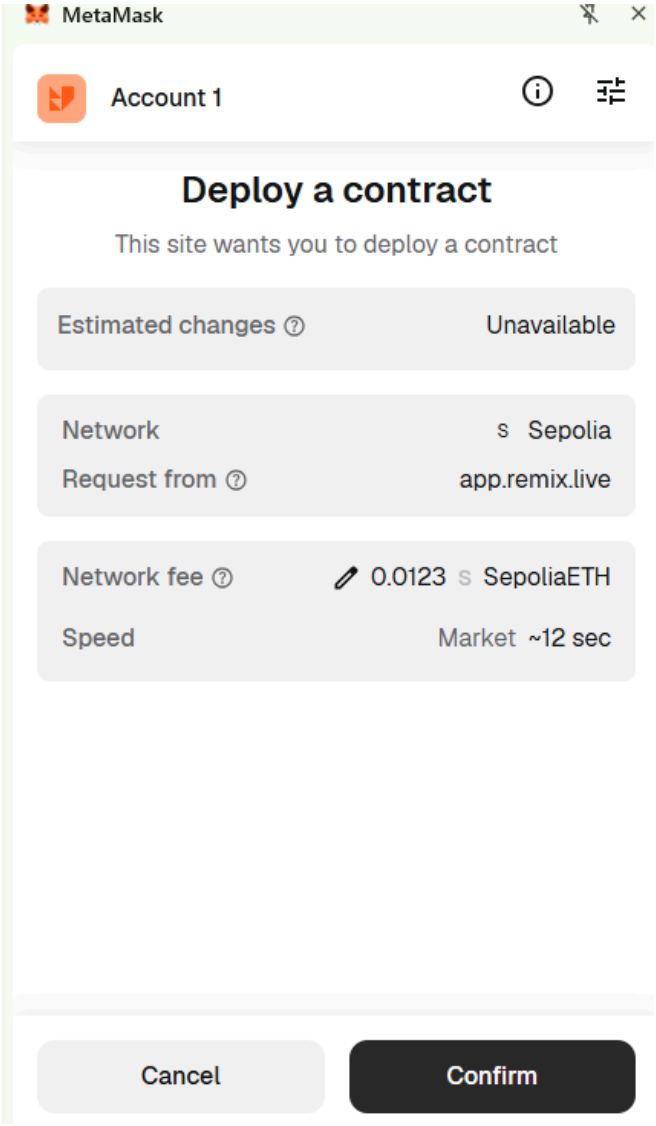

MetaMask
Account 1
0x1a9c2...3B827
4
0 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
TSM on the move
Share price falls despite Q3 earnings above estimates
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
S
S
-
—
0 SepoliaETH
Home
Perps
Swap
Activity

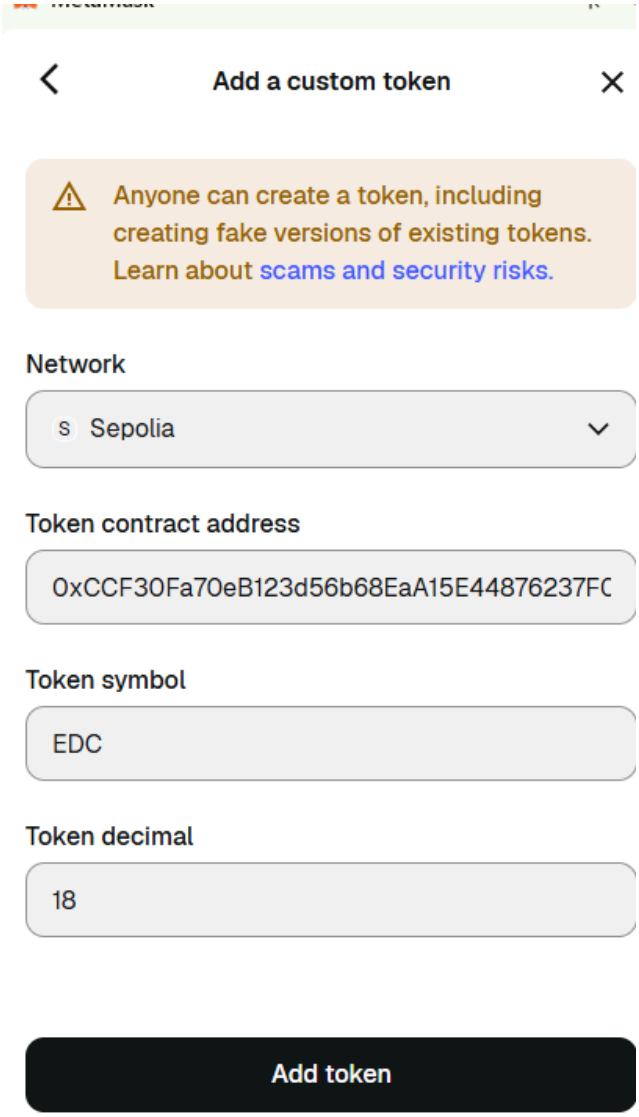

Ethereum Sepolia Faucet
BETA
Get free Sepolia ETH sent directly to your wallet. Brought to you by Google Cloud for Web3.
Get 0.05 Sepolia ETH
Transaction complete! Check your wallet address
Network
Ethereum Sepolia
Recipient
0x1a9c2CB1c68397De67C4CADD6b1C5E2C0713B8...
Transaction hash
0xfff60a3764092e33077fd824cdad11aa5948612c6...

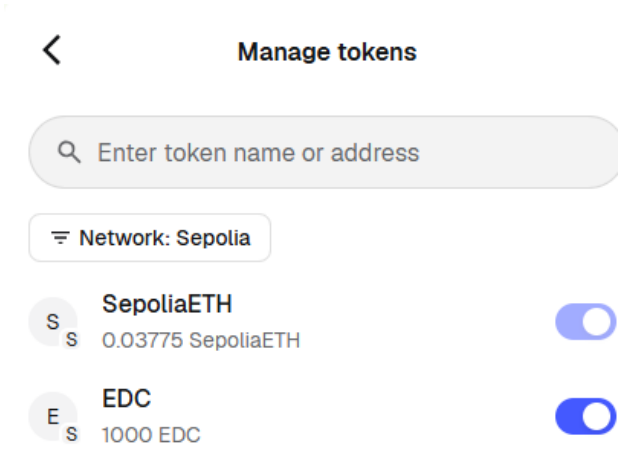

MetaMask
Account 1
0x1a9c2...3B827
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
TSM on the move
Share price falls despite Q3 earnings above estimates

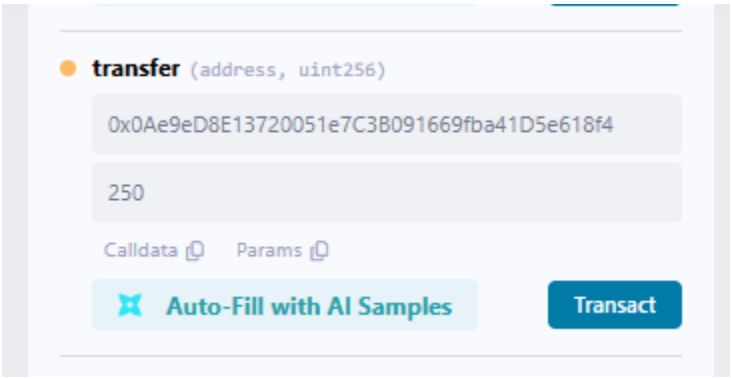
Tutorial 5:

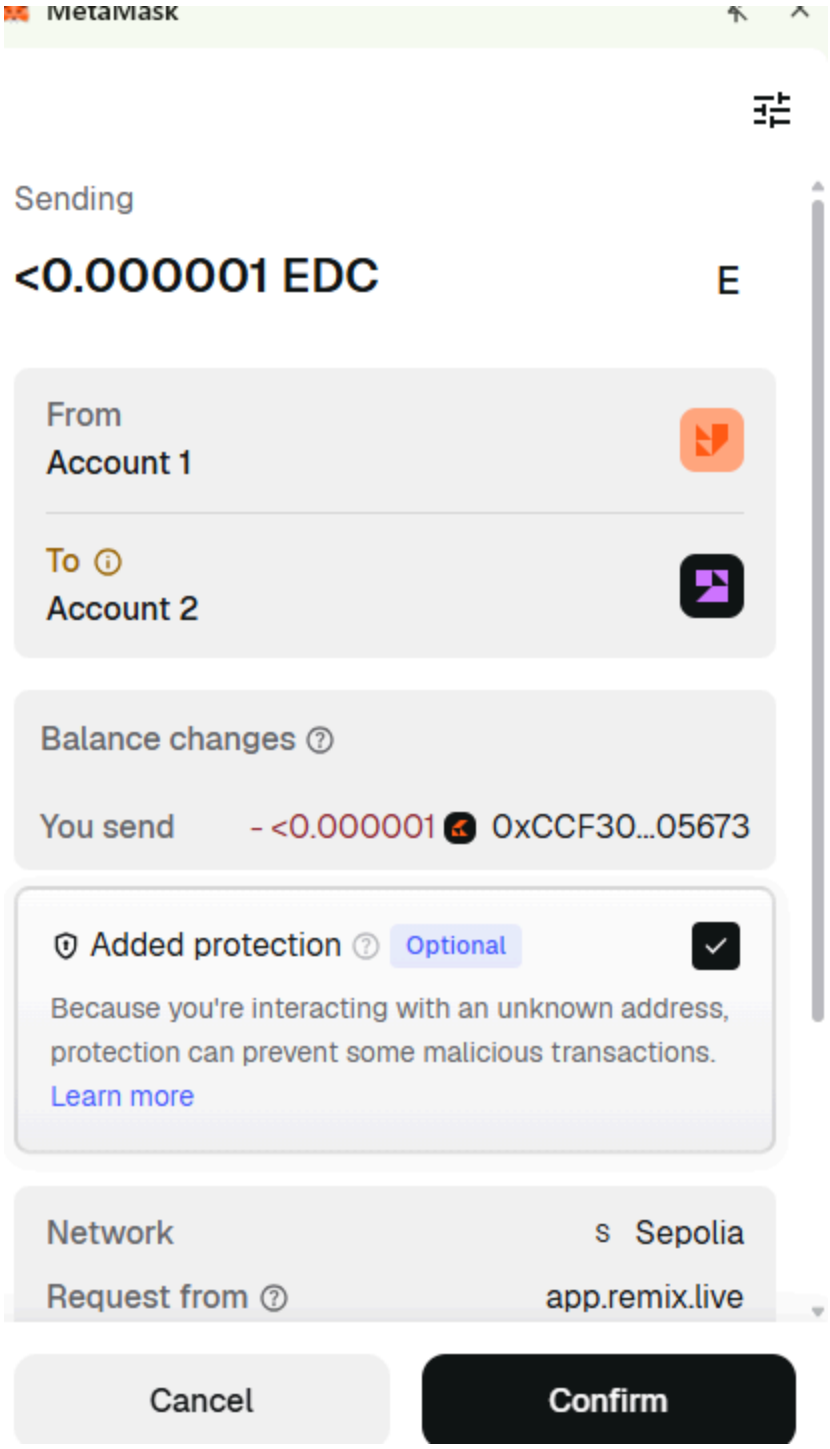

DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 16
0x1a9...3b827
0.050 ETH
Deploy
Deployed Contracts
Transaction History
EEZ
Deployed Contracts 0
+ Add Contract
No contracts deployed yet.
Go to the Deploy tab to deploy your first contract.
Compiled
Settings
introduction.sol
erc20interface.sol
erc20TokenCrea
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;
import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";
contract EduCoin is ERC20 {
constructor() ERC20("EduCoin", "EDC") { infinite gas 968200 gas
_mint(msg.sender, 1000 * 10 ** decimals());
}
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: EduCoin.decimals() data: 0x313...ce567
Debug
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: EduCoin.totalSupply() data: 0x181...60ddd
Debug
[vm] from: 0x5B3...eddC4 to: EduCoin.(constructor) value: 0 wei data: 0x608...20033 logs: 1 hash: 0x187...5c830
Debug
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
TSM on the move
Share price falls despite Q3 earnings above estimates
Tokens
DeFi
NFTs
Back up your Secret Recovery Phrase to keep your wallet and funds secure.
Back up now
app.remix.live
Account 1

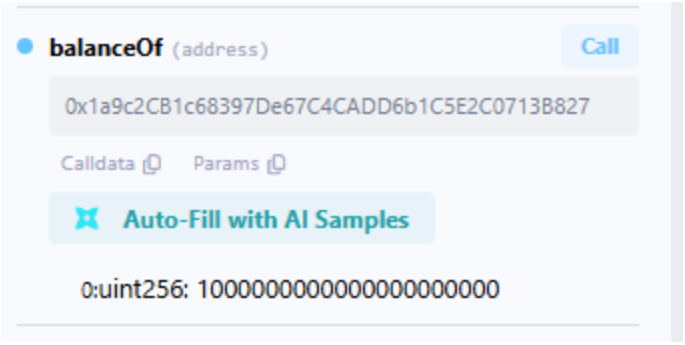

MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
Unavailable
Network
Sepolia
Request from
app.remix.live
Network fee
0.0123 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm

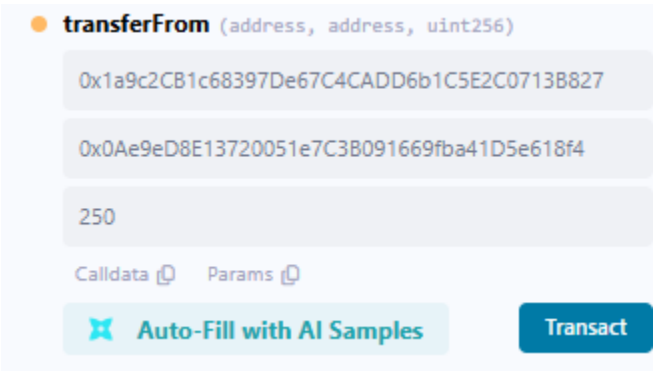

Add a custom token
Anyone can create a token, including creating fake versions of existing tokens. Learn about scams and security risks.
Network
Sepolia
Token contract address
0xCCF30Fa70eB123d56b68EaA15E44876237FC
Token symbol
EDC
Token decimal
18
Add token

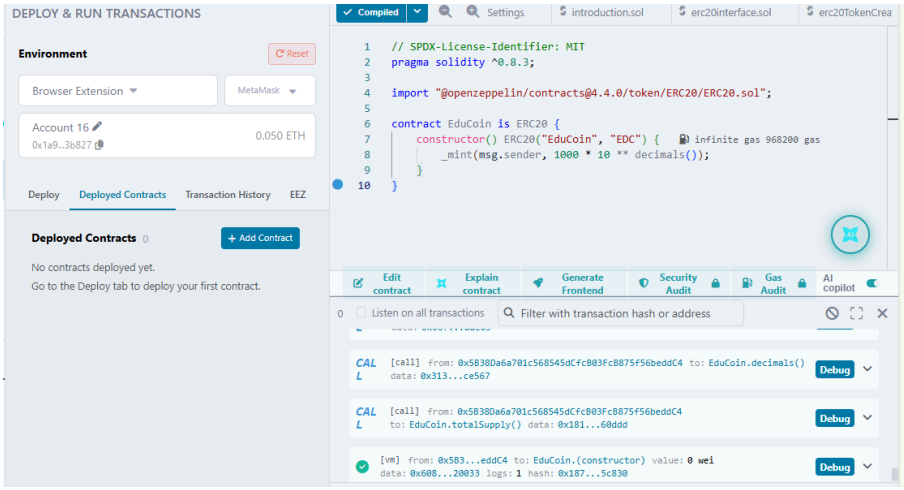

Manage tokens
Enter token name or address
Network: Sepolia
SepoliaETH
0.03775 SepoliaETH
EDC
1000 EDC

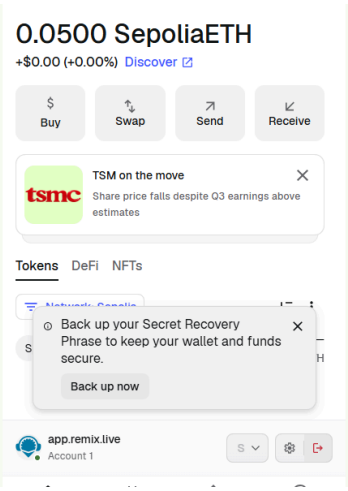

transfer (address, uint256)
0x0Ae9eD8E13720051e7C3B091669fba41D5e618f4
250
Calldata
Params
Auto-Fill with AI Samples
Transact

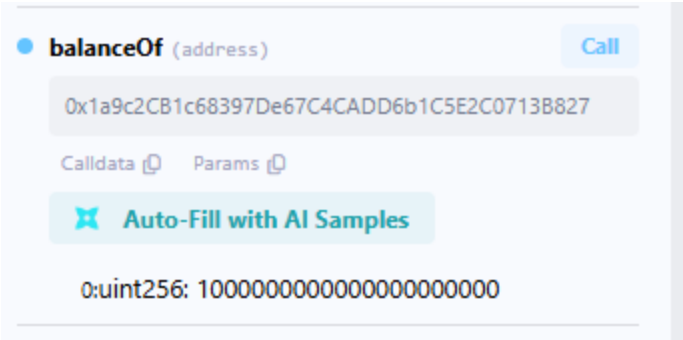

MetaMask
Sending
<0.000001 EDC
E
From
Account 1
To
Account 2
Balance changes
You send
- <0.000001
0xCCF30...05673
Added protection
Optional
Because you're interacting with an unknown address, protection can prevent some malicious transactions.
Learn more
Network
Sepolia
Request from
app.remix.live
Cancel
Confirm

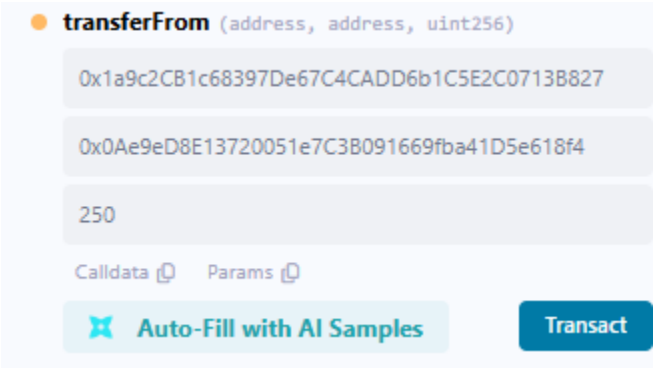

balanceOf (address)
Call
0x1a9c2CB1c68397De67C4CADD6b1C5E2C0713B827
Calldata
Params
Auto-Fill with AI Samples
0:uint256: 10000000000000000000


transferFrom (address, address, uint256)
0x1a9c2CB1c68397De67C4CADD6b1C5E2C0713B827
0x0Ae9eD8E13720051e7C3B091669fba41D5e618f4
250
Calldata
Params
Auto-Fill with AI Samples
Transact


DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 16
0x1a9...3b827
0.050 ETH
Deploy
Deployed Contracts
Transaction History
EEZ
Deployed Contracts 0
+ Add Contract
No contracts deployed yet.
Go to the Deploy tab to deploy your first contract.
Compiled
Settings
introduction.sol
erc20interface.sol
erc20TokenCrea
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;
import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";
contract EduCoin is ERC20 {
constructor() ERC20("EduCoin", "EDC") { infinite gas 968200 gas
_mint(msg.sender, 1000 * 10 ** decimals());
}
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: EduCoin.decimals() data: 0x313...ce567
Debug
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: EduCoin.totalSupply() data: 0x181...60ddd
Debug
[vm] from: 0x5B3...eddC4 to: EduCoin.(constructor) value: 0 wei data: 0x608...20033 logs: 1 hash: 0x187...5c830
Debug


0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
TSM on the move
Share price falls despite Q3 earnings above estimates
Tokens
DeFi
NFTs
Back up your Secret Recovery Phrase to keep your wallet and funds secure.
Back up now
app.remix.live
Account 1





Tutorial 6:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.2;

import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";
import "@openzeppelin/contracts@4.4.0/token/ERC20/extensions/ERC20Burnable.sol";
import "@openzeppelin/contracts@4.4.0/security/Pausable.sol";
import "@openzeppelin/contracts@4.4.0/access/Ownable.sol";

contract EduCoin is ERC20, ERC20Burnable, Pausable, Ownable {
    constructor() ERC20("EduCoin", "EDC") {    infinite gas 1738600 gas
        _mint(msg.sender, 1000 * 10 ** decimals());
    }

    function pause() public onlyOwner {    infinite gas
        _pause();
    }

    function unpause() public onlyOwner {    infinite gas
        _unpause();
    }

    function mint(address to, uint256 amount) public onlyOwner {    infinite gas
        _mint(to, amount);
    }

    function _beforeTokenTransfer(address from, address to, uint256 amount)    infinite gas
        internal
        whenNotPaused
        override
```

```
30      {
31          super._beforeTokenTransfer(from, to, amount);
32      }
33  }
```

Conclusion

Thus, the ERC-20 token EduCoin was deployed on the Ethereum Sepolia test network using Remix IDE and MetaMask. The experiment demonstrated the role of smart contracts in creating and managing fungible tokens, the importance of standard ERC-20 functions and events, and the process of connecting a wallet to a blockchain network. It also illustrated how token balances can be checked and tokens can be transferred between accounts. The experiment provided practical knowledge of ERC-20 token deployment and testing in a blockchain environment without using real cryptocurrency.